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Department of Computer Science and Engineering

FUNDAMENTALS OF DATA SCIENCE

CASE STUDY REPORT

Case Study 2: Customer Segmentation Using Clustering Analysis

Case Study 5: Wine Quality Assessment Using Physicochemical Properties

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Both case studies are done in one Jupyter Notebook (case_studies_2_and_5.ipynb) and are submitted through one GitHub repository.

Part A: Case Study 2 - Clustering Analysis

1. Problem Statement

East-West Airlines wants to understand its customers better. The company has data on how customers fly, how many frequent flyer miles they earn and use, and how they use the airline credit card. The task is to group similar customers into segments using clustering, so that the airline can plan offers and rewards for each group.

Three clustering methods are used and compared: K-Means, Hierarchical clustering and DBSCAN. The data has 3,999 customers and 12 columns, with no missing values.

2. Methodology

2.1 Data Preprocessing

- Missing values: none were found, so no filling was needed.
- Duplicates: one duplicate customer record was removed, leaving 3,998 customers.
- Columns used: the ID number was dropped because it is only a label. The three credit card columns (cc1, cc2, cc3) were dropped because they are category codes and not real amounts. The Award column was kept aside and used only to describe the clusters. Seven columns were used for clustering: Balance, Qual_miles, Bonus_miles, Bonus_trans, Flight_miles_12mo, Flight_trans_12 and Days_since_enroll.
- Outliers: a customer was removed if any of the seven values was more than 5 standard deviations away from the average. This removed 125 customers and 3,873 remained.
- Scaling: all seven columns were standardised (mean 0, standard deviation 1) so that large numbers such as Balance do not dominate small ones such as flight transactions.

2.2 Algorithms and Settings

- K-Means: K was tried from 2 to 10. The elbow curve and the silhouette score were used to choose K.
- Hierarchical clustering: a dendrogram was drawn and four linkage types (Ward, complete, average and single) were compared.
- DBSCAN: a k-distance plot was drawn, then 15 combinations of eps (0.5 to 2.5) and min_samples (5, 10, 20) were tested.
- Evaluation: the silhouette score (higher is better) and the Davies-Bouldin index (lower is better).

3. Statistical Findings

3.1 Distribution of the Features

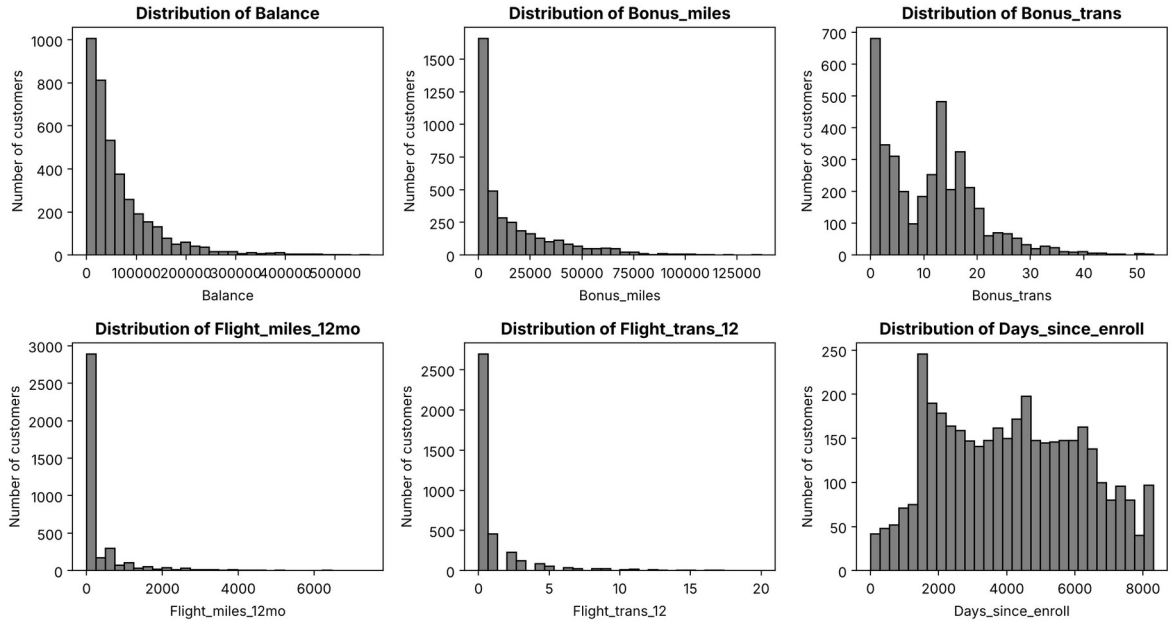


Figure 1: Histograms of six customer features (after cleaning)

Figure 1 shows that most features are not evenly spread. Balance, Bonus_miles, Flight_miles_12mo and Flight_trans_12 have a long tail on the right. Their skewness values are 2.46, 1.95, 3.98 and 3.71, which means most customers have small values and a few customers have very large values. Flight_miles_12mo and Flight_trans_12 have a very tall bar at zero, so a large part of the customers did not fly at all in the last 12 months. Days_since_enroll is almost balanced (skewness 0.12), which shows customers joined steadily over the years.

3.2 Relationship Between Features

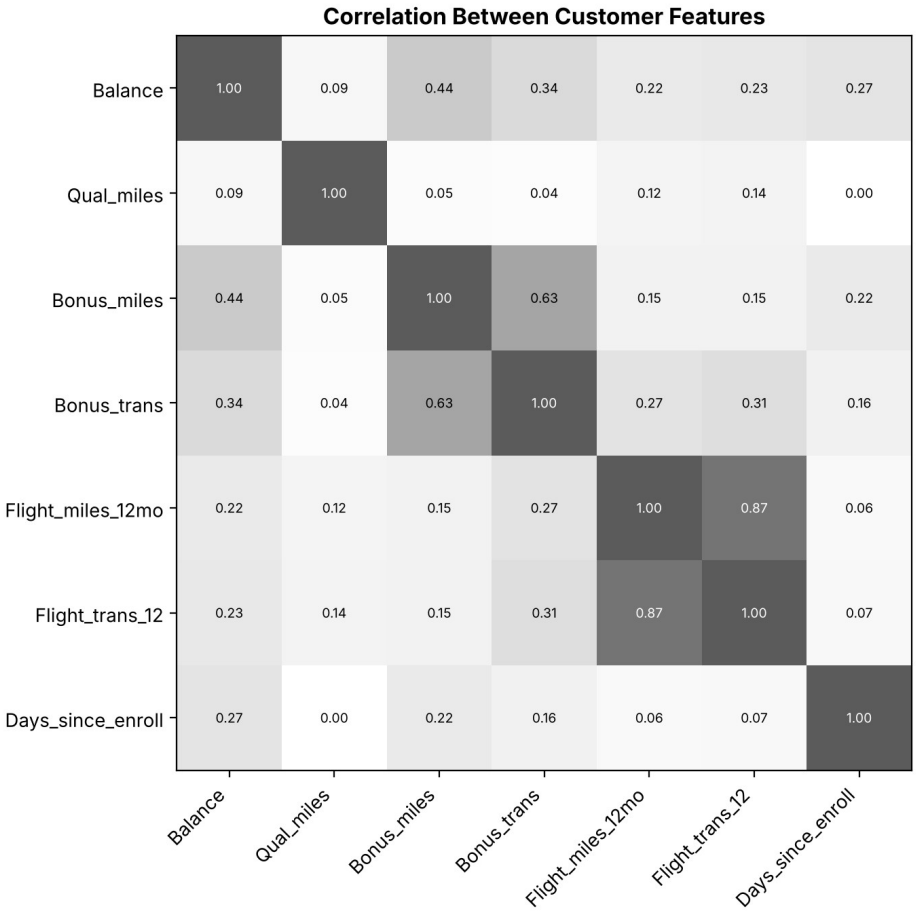


Figure 2: Correlation heatmap of the seven clustering features

In Figure 2, darker shades show a stronger relationship and the printed number shows the direction. Flight_miles_12mo and Flight_trans_12 are strongly related (0.87), which is natural because more flights give more miles. Bonus_miles and Bonus_trans are moderately related (0.63), and Balance has a moderate link with Bonus_miles (0.44). Qual_miles has almost no relation with the other columns (all values 0.14 or below), so it describes a different kind of customer behaviour.

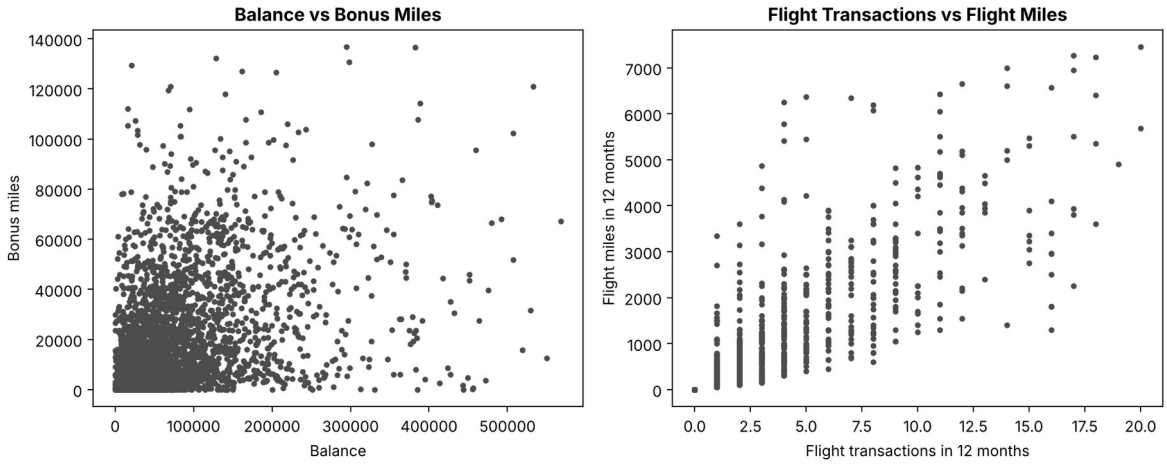


Figure 3: Balance against bonus miles, and flight transactions against flight miles

The left plot of Figure 3 shows that customers with a high balance usually have more bonus miles, but the points are widely spread. The right plot shows an almost straight line, which agrees with the 0.87 correlation. A big group of points sits at the bottom left, meaning many customers have little activity. These patterns suggest that clear groups of customers exist.

4. Model Evaluation

4.1 K-Means

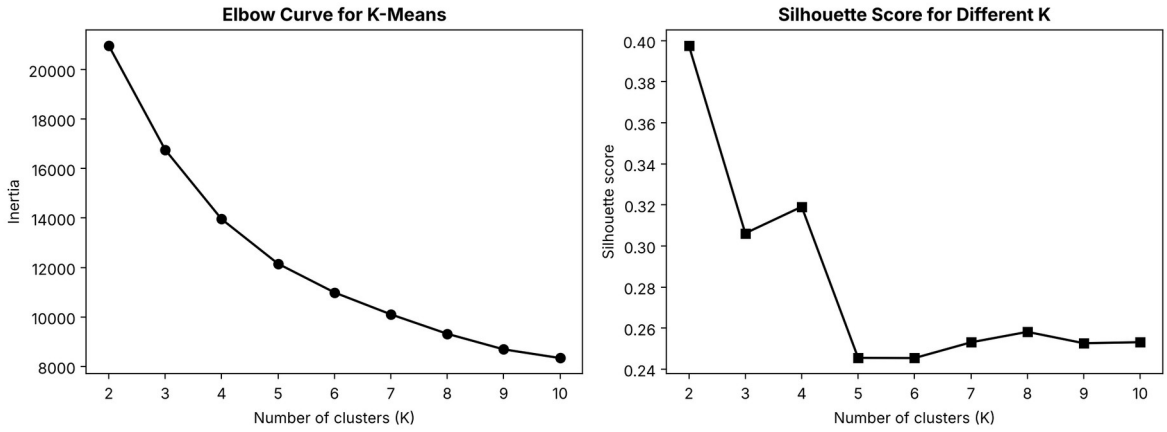


Figure 4: Elbow curve and silhouette score for K from 2 to 10

Table 1: K-Means results for different values of K

K	Inertia	Silhouette
2	20,974	0.398
3	16,763	0.306
4	13,956	0.319
5	12,153	0.245
6	10,991	0.245
7	10,105	0.253
8	9,319	0.258

The elbow curve in Figure 4 bends clearly around K = 4, after which the fall in inertia becomes slow. The silhouette score is highest at K = 2 (0.398), but two groups are too few to be useful for marketing. K = 4 has the second best score (0.319) and gives groups that are easy to explain, so K = 4 was selected.

4.2 Hierarchical Clustering

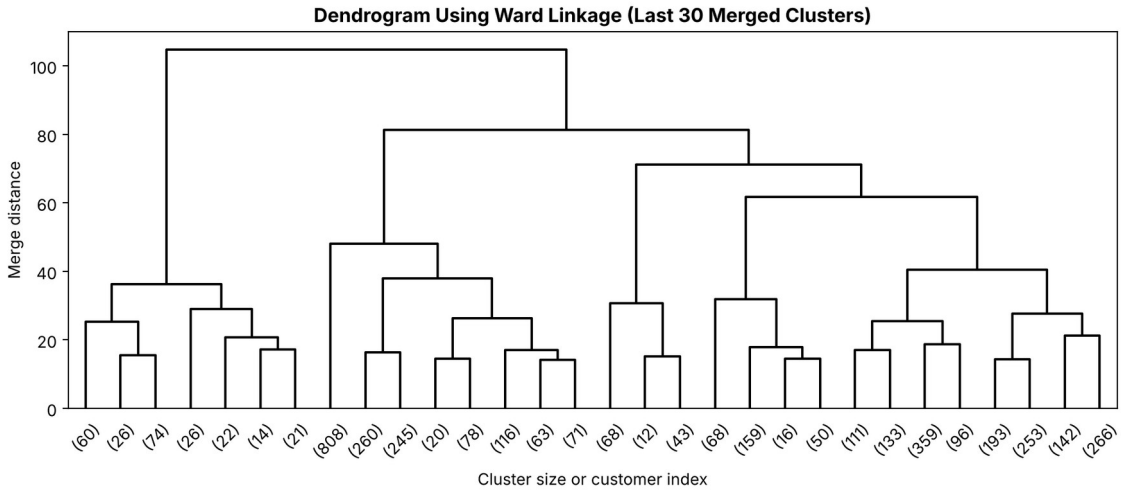


Figure 5: Dendrogram using Ward linkage (last 30 merged clusters)

The dendrogram in Figure 5 shows that the biggest jump in merge distance is at the top, where two large groups join, and cutting it a little lower gives four clusters. Table 2 compares the four linkage types for four clusters.

Table 2: Comparison of linkage types (4 clusters)

Linkage	Silhouette	Davies-Bouldin	Cluster sizes
Ward	0.211	1.339	1846, 1661, 243, 123
Complete	0.593	1.091	3729, 67, 56, 21
Average	0.580	0.983	3745, 116, 10, 2
Single	0.598	0.237	3870, 1, 1, 1

Complete, average and single linkage show high silhouette scores, but this is misleading. They put almost every customer in one huge cluster and leave tiny clusters with only a few customers (single linkage has three clusters with just one customer each). Such results are not useful. Ward linkage gives four balanced and meaningful groups even though its score is lower (0.211), so Ward was selected.

4.3 DBSCAN

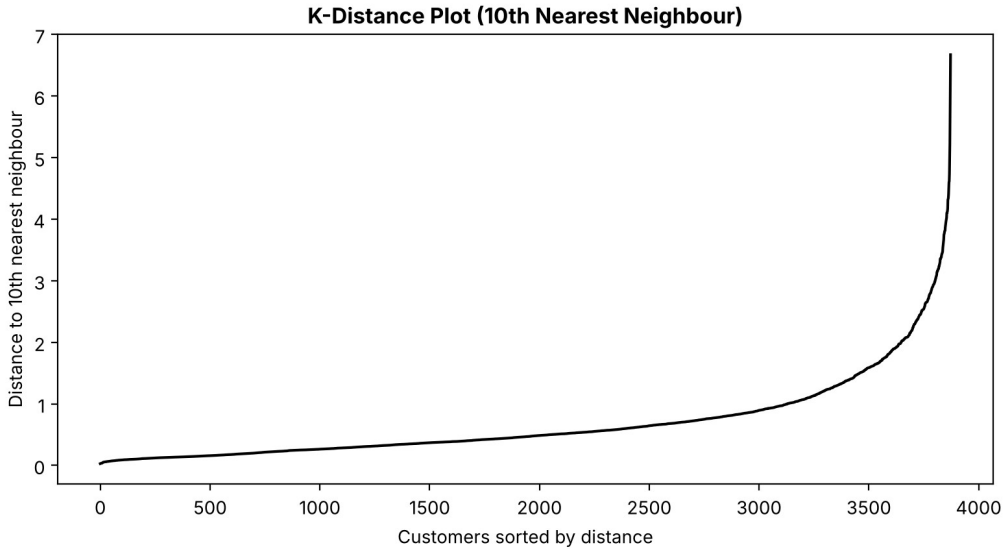


Figure 6: K-distance plot using the 10th nearest neighbour

The k-distance plot in Figure 6 is flat for most customers and rises sharply near the end. The bend is between 1 and 2, so eps values around this range were tested. The table of 15 combinations showed that very small eps (0.5) marks too many customers as noise (1,235 to 1,734), and larger eps merges nearly everything into one cluster. The best setting that gave at least two clusters with noise below 10% of the data was eps = 1.5 and min_samples = 10.

With these settings DBSCAN found two clusters and 269 noise customers (about 7%). One cluster holds 3,579 customers and the other holds only 25 customers. The noise customers are unusual: their average balance is about 163,900 and their average flight miles are about 2,179, which is much higher than the main cluster (about 59,557 and 205). DBSCAN therefore is good at finding exceptional customers, but it does not split the normal customers into segments.

4.4 Visual Comparison of the Clusters

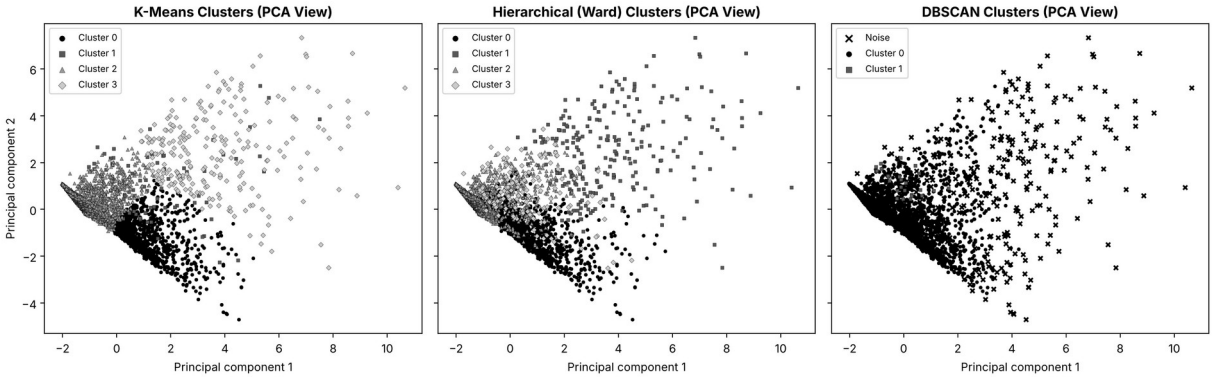


Figure 7: Clusters from the three methods shown on two principal components

To draw the seven-dimensional data on a flat page, PCA was used. The first two components keep 57.3% of the information (36.5% and 20.8%). In Figure 7, the K-Means and Hierarchical plots show groups lying in different areas of the plot, with the active customers spread to the upper right. Some overlap is visible, which explains the modest silhouette scores. In the DBSCAN plot, the cross marks (noise) lie at the outer edge and nearly all other customers form a single mass.

4.5 Meaning of the K-Means Clusters

Table 3: Average values of each K-Means cluster

Cluster	Customers	Balance	Qual miles	Bonus miles	Flight miles	Days enrolled	Award rate
0	1,157	117,369	13	36,598	210	5,126	0.52
1	87	97,916	2,384	19,639	615	4,184	0.62
2	2,386	35,667	10	5,063	116	3,568	0.24
3	243	118,820	126	25,374	3,081	4,421	0.65

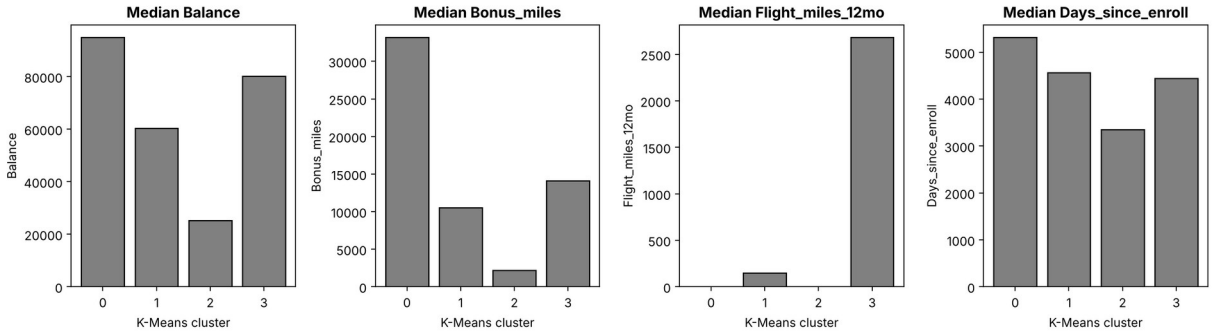


Figure 8: Median values of key features for each K-Means cluster

- Cluster 0 - Loyal reward earners (1,157 customers): the longest members (about 5,126 days), high balance and many bonus miles, but they hardly fly. They earn miles mostly through the credit card and other partners.

- Cluster 1 - Status chasers (87 customers): very high qualifying miles (about 2,384), a small group that works hard to reach the airline's membership levels.
- Cluster 2 - Low-activity customers (2,386 customers): the largest group, with the lowest balance, bonus miles and flights, and the newest members. Only 24% have taken an award, so there is large room to increase their engagement.
- Cluster 3 - Frequent flyers (243 customers): about 3,081 flight miles and almost 9 flights in 12 months, with a high balance and the highest award rate (0.65). These are the most valuable travellers.

The Hierarchical (Ward) clusters show a similar story: a large low-activity group (1,661), a large medium group (1,846), a frequent flyer group (243) and a small group with high qualifying miles (123). This agreement between two different methods gives more confidence that these segments are real.

4.6 Summary of Metrics

Table 4: Evaluation of the three methods

Method	Clusters	Noise points	Silhouette	Davies-Bouldin
K-Means	4	0	0.319	1.178
Hierarchical (Ward)	4	0	0.211	1.339
DBSCAN	2	269	0.617	0.515

DBSCAN has the best numbers, but its silhouette score was calculated without the 269 noise customers and almost all remaining customers are in one cluster, so the score is high only because the task became easy. K-Means has a better score than Ward on both measures and gives clearly different and useful segments, so it is the best method for this business problem.

5. Conclusion

Customers of East-West Airlines can be divided into four useful segments: loyal reward earners, status chasers, low-activity customers and frequent flyers. K-Means with K = 4 gave the best balance between a good score and groups that a business can act on. Hierarchical clustering with Ward linkage supported the same picture, and DBSCAN helped to identify about 7% of very unusual customers with very high balances and miles.

The airline can offer travel-based rewards to the frequent flyers, tier-upgrade offers to the status chasers, and special activation offers to the large low-activity group. The main limitation is that the groups overlap somewhat (silhouette about 0.32), because customer behaviour changes smoothly and does not form sharp groups.

Part B: Case Study 5 - Wine Quality Assessment

1. Problem Statement

The company now checks wine quality only through tasting by experts. Management wants to know whether laboratory measurements can explain why some wines get better scores than others. The goals are to find which measurements are most strongly linked to quality and to build a multiple linear regression model that predicts the quality score.

The data contains 1,599 red and 4,898 white Portuguese "Vinho Verde" wines, which makes 6,497 wines altogether. Each wine has 11 laboratory measurements and a quality score between 0 and 10 given by wine experts (the median of at least three experts). There are no missing values.

2. Methodology

- The red and white wine files were joined into one dataset of 6,497 wines.
- Task 1: mean, median, mode, standard deviation and skewness were found for alcohol, volatile acidity and residual sugar. The coefficient of variation (standard deviation divided by mean) was used to compare variability. Pearson correlation was used to study alcohol and quality.
- Task 2: a histogram, a box plot and a correlation heatmap were drawn.
- Task 3: the data was split 80% for training (5,197 wines) and 20% for testing (1,300 wines) with random_state = 42. A linear regression model was fitted on the training data and tested on the test data using MAE, MSE, RMSE and R-squared. Standardised coefficients and VIF values were used to study the predictors.

3. Statistical Findings

3.1 Task 1: Statistical Analysis

Table 5: Descriptive statistics of three attributes

Attribute	Mean	Median	Mode	Std Dev	Skewness	CV (%)
Alcohol	10.492	10.30	9.50	1.193	0.566	11.37
Volatile acidity	0.340	0.29	0.28	0.165	1.495	48.47
Residual sugar	5.443	3.00	2.00	4.758	1.435	87.41

For all three attributes the mean is larger than the median, and the median is larger than the mode, which is the pattern of a right-skewed distribution. Alcohol is only mildly skewed (0.566). Volatile acidity (1.495) and residual sugar (1.435) are strongly skewed, so most

wines have low values and a few wines have very high values. For example, the average residual sugar is 5.44 but the typical (median) wine has only 3.00.

The coefficient of variation shows that residual sugar has the greatest relative variability (87.41%), followed by volatile acidity (48.47%) and alcohol (11.37%). This measure is used because the three attributes have different units and sizes, and the standard deviation alone cannot be compared fairly.

The Pearson correlation between alcohol and quality is r = 0.444 with a p-value that is practically zero. The relationship is positive and moderate: wines with more alcohol tend to receive higher quality scores. Alcohol alone explains about 20% of the differences in quality (r squared is 0.197), so other factors also matter.

3.2 Task 2: Exploratory Data Analysis

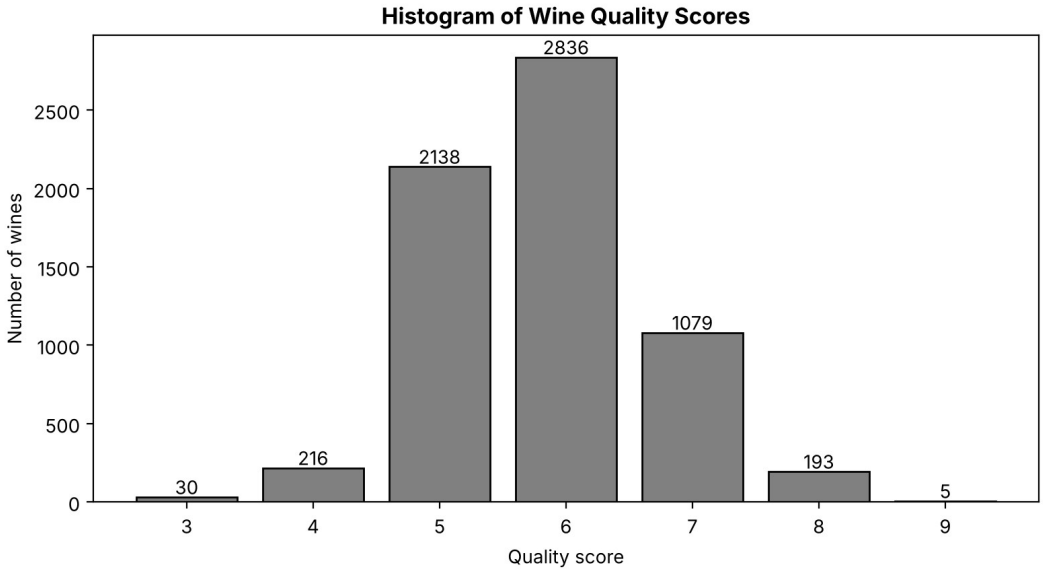


Figure 9: Histogram of wine quality scores

Most wines get an average score. Scores 5 and 6 together cover 76.6% of all wines (2,138 and 2,836 wines). Very poor wines (score 3) are only 30 and excellent wines (score 9) are only 5. The shape is close to a bell curve centred near 6, with a slightly longer tail towards high scores. Because the data is imbalanced at both ends, any model will find it difficult to learn what makes a wine very bad or very good.

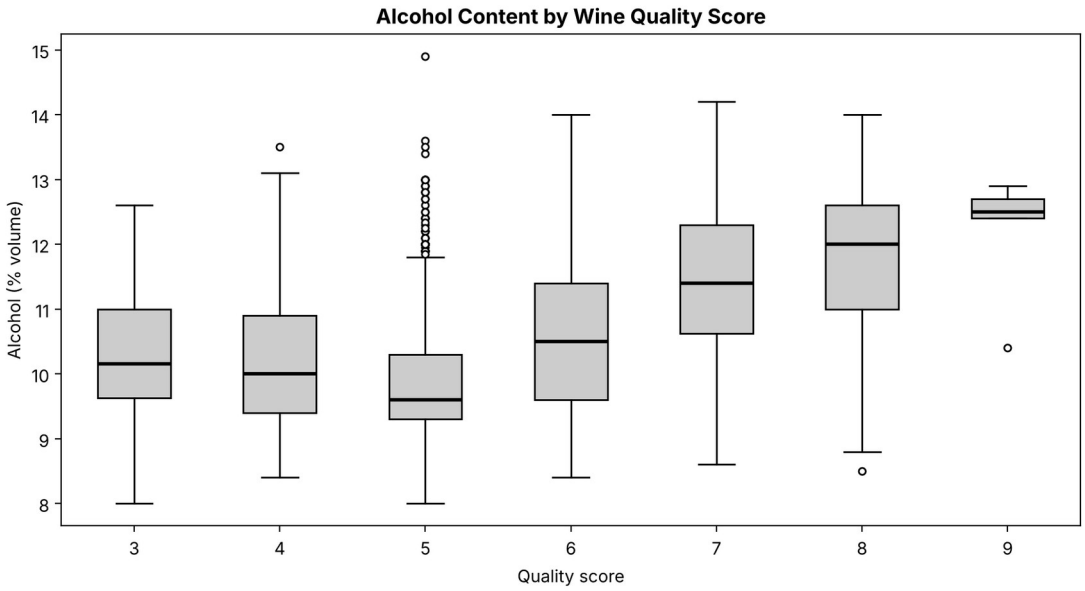


Figure 10: Box plot of alcohol content for each quality score

The box plot shows a clear rise in alcohol with quality from score 5 onwards. The median alcohol is 9.6% for quality 5, 10.5% for quality 6, 11.4% for quality 7, 12.0% for quality 8 and 12.5% for quality 9. Quality 3 and 4 wines have medians (10.15% and 10.0%) a little higher than quality 5 wines, so the relationship is not perfectly straight at the low end. Quality 5 has the most outliers (55 wines with unusually high alcohol), while quality 4, 8 and 9 have one outlier each.

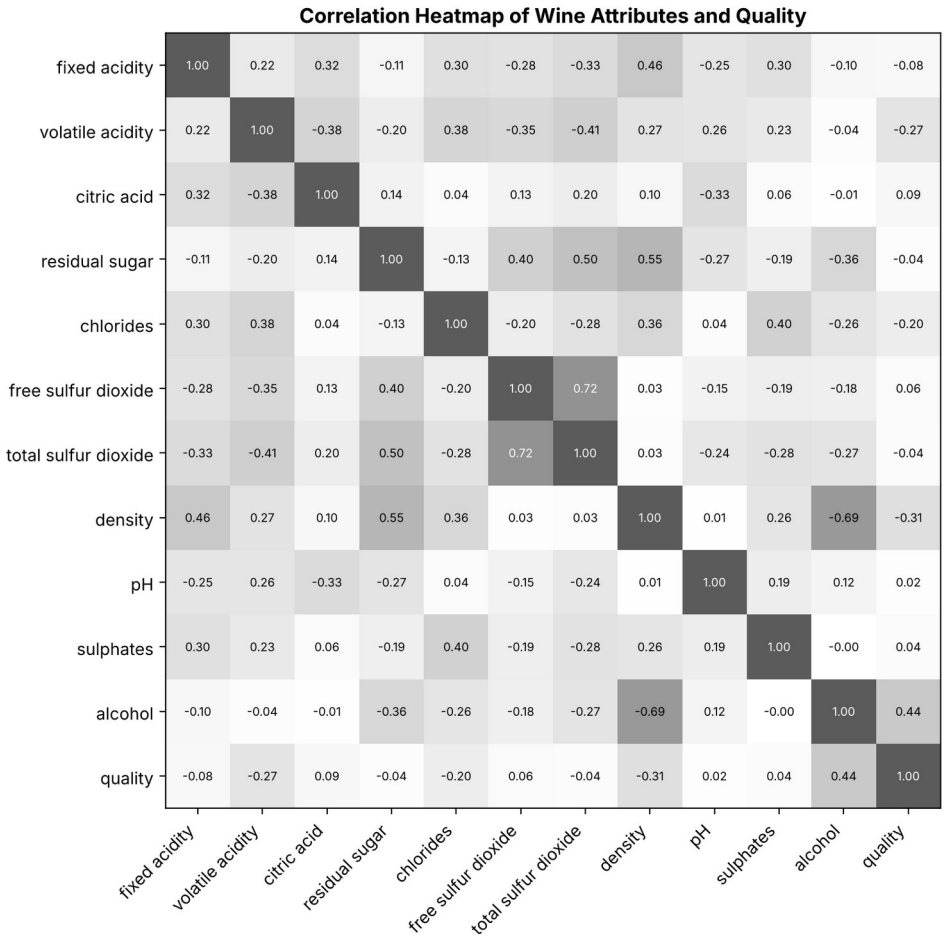


Figure 11: Correlation heatmap of the 11 attributes and quality

In Figure 11 darker shades mean a stronger relationship and the printed values show the direction. The attributes related most to quality are alcohol (0.44), density (-0.31), volatile acidity (-0.27) and chlorides (-0.20). The heatmap also shows strong links among the inputs: alcohol and density (-0.69), free and total sulfur dioxide (0.72), and residual sugar and density (0.55). This is an early warning of multicollinearity.

Based on absolute Pearson correlation, the two attributes most strongly associated with quality are alcohol (r = +0.444, positive, so more alcohol goes with better quality) and density (r = -0.306, negative, so denser wines tend to score lower). Density is probably not a direct cause. Alcohol is lighter than water, so wines with more alcohol have lower density, and density partly repeats the alcohol effect.

4. Model Evaluation

4.1 Test Set Performance

Table 6: Linear regression results on the test set (1,300 wines)

Measure	Value	Meaning
MAE	0.566	On average the prediction is off by about 0.57 points
MSE	0.547	Average squared error
RMSE	0.739	Typical error, giving more weight to big mistakes
R-squared	0.260	The model explains 26% of the differences in quality

A simple guess that always predicts the average training quality gives an RMSE of 0.860. The model reduces this to 0.739, which is an improvement of about 14%. So the model has learned something real, but the improvement is modest. When the predictions are rounded to the nearest whole score, 53.9% match the expert score exactly and 94.8% are within one point.

The model was also tried separately on each wine type. For red wine, R-squared was 0.403 (RMSE 0.625) and for white wine it was 0.265 (RMSE 0.754). The red wine model is better, which suggests that red and white wines behave differently and one common line is not ideal.

4.2 Influential Predictors and Multicollinearity

Table 7: Standardised coefficients and VIF values of the model

Attribute	Standardised coefficient	VIF
Alcohol	+0.322	4.62
Residual sugar	+0.220	7.20
Volatile acidity	-0.219	1.94
Density	-0.177	15.51
Total sulfur dioxide	-0.150	2.96
Free sulfur dioxide	+0.123	2.16
Sulphates	+0.120	1.54
Fixed acidity	+0.102	4.74
pH	+0.077	2.47
Citric acid	-0.021	1.61
Chlorides	-0.012	1.65

The two most influential predictors by size of standardised coefficient are alcohol (+0.322) and residual sugar (+0.220). A change of one standard deviation in alcohol changes the predicted quality by 0.32 points. Volatile acidity (-0.219) is almost as strong as residual sugar, so the second place is very close.

The VIF values must be read before trusting these numbers. Density has a VIF of 15.51, which is very high (values above 10 are a concern), and residual sugar has 7.20. These attributes move together with alcohol and with each other, so the model cannot separate their effects clearly. This explains a confusing result: residual sugar has almost no direct correlation with quality (-0.037), yet it gets a positive coefficient in the model. Its sign appears because density is also in the model, and the two effects are mixed. The effect of residual sugar and density should therefore not be interpreted on its own. Alcohol (VIF 4.62) and volatile acidity (VIF 1.94) are more reliable because their VIF values are lower and their signs match their correlations.

4.3 Predicted and Actual Scores

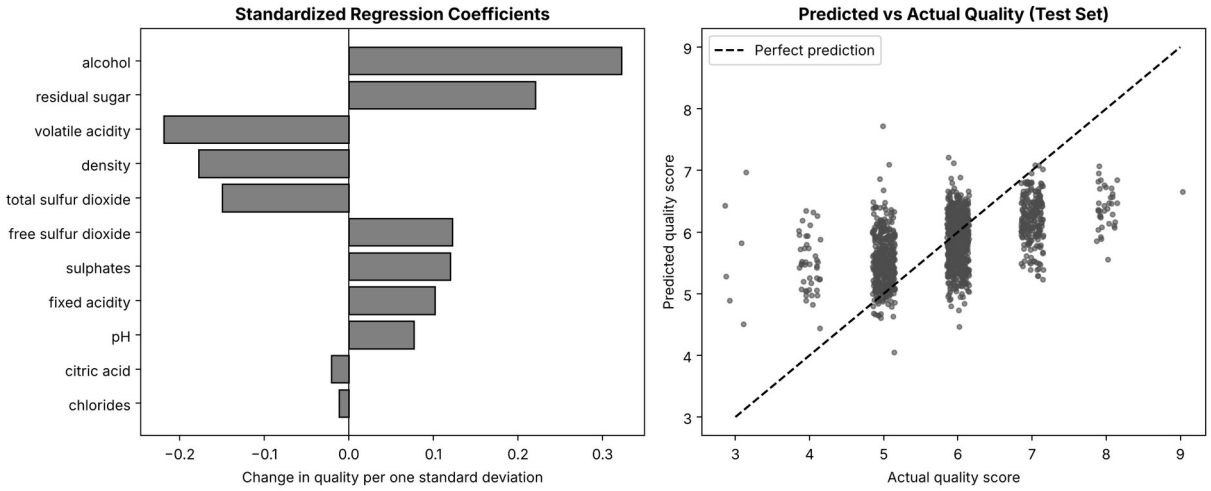


Figure 12: Standardised coefficients (left) and predicted against actual quality on the test set (right)

If the model were perfect, all points would lie on the dashed line. In the right plot the points form a wide cloud instead. The predictions are squeezed into the range of about 5 to 7, even when the actual score is 3, 4, 8 or 9. Average wines (5 and 6) are predicted reasonably, but poor wines are predicted too high and excellent wines are predicted too low. Limitations of the model are therefore:

- A straight-line model cannot capture complex or curved relationships between measurements and taste.
- The scores are whole numbers from subjective tasting, and few wines exist at the extreme scores, so the model learns mostly about average wines.
- Strongly related inputs (density, alcohol, sugar) make individual coefficients unstable.
- Red and white wines were combined in one model, although they behave differently.

5. Business Decision-Making

5.1 Measurements to Focus On During Screening

- Alcohol: it has the strongest positive link with quality (r = 0.44) and the largest coefficient in the model, and the box plot shows quality rising steadily with alcohol.
- Volatile acidity: it has a clear negative link with quality (r = -0.27), a strong coefficient (-0.219) and a low VIF (1.94), so its effect is stable and trustworthy. Density was not chosen even though its correlation is higher, because it mostly repeats the alcohol information and has a high VIF.

5.2 Can the Model Replace Sensory Evaluation?

No. The model is not reliable enough to replace tasting by experts. The R-squared on the test set is only 0.26, so about 74% of the differences in quality remain unexplained. The RMSE of 0.739 and MAE of 0.566 are large compared with a quality scale where most wines lie between 5 and 7, and only 53.9% of rounded predictions match the expert score exactly. The model also fails at the two ends of the scale, which are the poor and excellent wines that the company cares about most. It is better used as a low-cost first check that flags wines for closer tasting.

5.3 Limitations

- The quality scores come from human tasters, so they are subjective, and the dataset has no information about grape type, brand or price. Extreme scores have very few examples.
- The analysis uses only a linear model with one train-test split, and joins red and white wine together. Non-linear models and separate models for each wine type could give better results.

6. Conclusion

Alcohol is the most important laboratory measurement linked to wine quality, followed by density and volatile acidity. The linear regression model gives an R-squared of 0.26, an RMSE of 0.739 and an MAE of 0.566 on the test set. It is better than a simple average guess but not accurate enough to replace expert tasting. The company can use alcohol and volatile acidity as quick screening measures, while keeping sensory evaluation for the final decision.

References

P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis, "Modeling wine preferences by data mining from physicochemical properties," Decision Support Systems, Elsevier, 47(4):547-553, 2009.

East-West Airlines customer dataset (EastWestAirlines.xlsx) provided for the case study.

Scikit-learn, SciPy, pandas and matplotlib documentation (Python libraries used for the analysis).